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SET-I

Answer 1

HTML is the core thing you need if you want to create a page on the internet. Its full name is Hyper Text Markup Language. One major point to remember is that it is not a real programming language like C graphics or Java where you write algorithms, it is just a simple tag language to arrange items. HTML works by giving simple commands to browsers like chrome so it understands where to display paragraphs, normal photos, table data, and links on a laptop screen. If you ask me, if you want to learn web development, you have to start with HTML files because it builds the raw skeleton. It is super simple to understand so every BCA first-year student can learn it fast.

Coming to features, HTML is totally free and it opens nicely on any old browser whether it is chrome or firefox. It does not look at your laptop operating system, meaning it runs exactly same on windows or mac. Another feature is we can put heavy media files inside it like adding mp3 songs, videos, and normal images easily. If we combine HTML with styling and basic script, we can make moving websites. The main use of HTML is to throw your raw content in proper places on the webpage. Every single running website has some HTML tags inside it. We also use it for creating email templates and login screen layouts.

If you see the history, HTML has changed a lot over time. The very first version was HTML 1.0 which had very less tags and you can only make plain text pages without layout. Then version 2 and 3 brought table tags and text bold features. For a long time, people used version 4.01 which was okay for forms. But now, we use HTML5 everywhere. This new version is awesome because it has direct tags to play audio and video so no extra media player software is needed. It also has easy tags like header, footer, and section so the browser knows what block is kept where.

The code structure is very tiny. At the first line, we write a tag to show it is the new version. Then we open html and close it at the last line. The whole page has two main zones: head part and body part. The head section uses head tag. Whatever you type here never shows up on the main screen to website visitors. It is for background stuff, like the tab title at the top or linking design sheets. The main active area is the body section. Whatever paragraph, image, button, or form you see on a live page goes inside this body block only. Previously, websites were designed only for big PC monitors. But now, everyone uses smartphones. So HTML5 has special viewport tags that tell phones to scale down the text automatically so layout doesn't look broken or tiny.

Answer 2

CSS basically stands for Cascading Style Sheets. We need it because HTML alone looks super ugly and dry, like a plain notepad file. HTML only puts raw text on the screen, but CSS adds color, styling, custom fonts, and professional layouts. We can change sizes, shift boxes left or right, and clear up gaps between paragraphs. Without CSS, your college website will look like a boring old text document.

We can add CSS styling in exactly three ways. First is Inline CSS where you write design inside the HTML tag itself using style attribute. Like writing style inside an h1 tag. This is okay for small testing of one single line but makes code messy for big projects. Second is Internal CSS where we put a style tag block inside the head zone of that page. It works fine if you want to design just one single webpage. Third is External CSS which is the best way used by programmers. We make a separate blank file with a dot css name, write all design rules there, and connect it using a link tag in HTML. The biggest benefit is you can link this one css file to 20 different pages. If you want to change background from black to white, you just change one line in that css file and the whole website updates instantly.

CSS helps us build clean layouts by breaking the page into headers, sidebars, main content boxes, and footers. For mobile screens, it has media queries. This means we can write logic like: if the user screen is small like mobile phone, change the text font and stack the menu vertically instead of horizontal. To link these styles to text, we use selectors called classes and IDs. A Class is used when you want multiple things to look same, and it is written with a dot symbol like dot myDesign in css. You can repeat class names. An ID is unique and can be used only once per page for a main element, and we write it with a hash symbol.

Common issues in CSS are broken layouts or text overlapping on top of each other. This happens due to missing semicolons, wrong spelling of classes, or styles fighting with each other in the background. You can fix this by right-clicking the page and opening Inspect in chrome browser. It highlights the wrong line directly so you can edit it.

Answer 3

Tables are used in HTML when we want to show numbers or data in a clean row and column format. It makes data very easy to see for the user. For example, if you want to show a student mark sheet, or college exam timetable, or price list, tables are the best choice. To make a table in HTML, we use the table tag. Inside this, we create horizontal rows one by one using tr tag which means table row. Inside each row, we put our data cells. If it is the first top row with column titles, we use th tag which means Table Heading. The benefit of th is that the browser automatically makes that text bold and center so it looks different. For all other normal data cells below, we use td tag which means Table Data. We can put many columns inside a row, but we must make sure every row has same number of cells otherwise table shape will look broken. Remember, tables are only for data list, never use them for making full website design layout because that is bad practice.

Apart from tables, HTML lists are also there to group items together so user does not get confused by long paragraphs. There are three types of lists we can make. First is Ordered List which uses ol tag. This list automatically numbers the items like 1, 2, 3 or A, B, C. We use this when order is important, like writing steps to install some software or recipes. Second is Unordered List which uses ul tag. This just puts simple bullet points or dots before the items.

It is used when order does not matter, like a list of books or shopping items. For both lists, the items inside are written using li tag. The third type is Description List which uses dl tag. It doesn't use bullets, it is used like a dictionary where you have a term using dt and then its meaning using dd below it.

To make our text content look neat and readable, we use text formatting tags. If we want to make some words look dark and bold because they are important, we use b or strong tag. If we want to tilt text for notes, we use i or em tag for italics. To draw a line under text, we use u tag. We also use br to break the line and hr to draw a horizontal line. When making tables and lists, we must follow some rules so that blind people using screen readers can use our site. We should use caption tag for table titles and proper th headings so software can read it properly.

SET-II

Answer 4

HTML forms are the main thing if you want to make a website interactive. Without forms, a website is just a dead page where users can only read text but cannot do anything back. Forms allow users to type data, upload files, and send information directly to the web server database. For example, when you fill college admission form online, or login to your email account, or type something in google search box, you are using an HTML form only.

Every form in HTML must be written inside opening form and closing form tags. This tag works like a box for all our inputs and buttons. The form tag has two main attributes which are very important. First is action attribute. This takes the name of the backend file where our data should go for saving. Second is method attribute, which decides how data is sent. It can be GET or POST. GET method is very simple and shows all typed data inside the top URL address bar, so it is not safe for passwords because anyone can see it. POST method hides all data in the background packet, so we always use POST for login forms so passwords stay secret.

Inside the form, the most common tag is input tag. This tag does not need a closing tag, and by changing its type attribute, we can make different things. If we write type text, it makes a normal box for typing names. If we change it to type password, the browser automatically hides the letters and shows black dots so nobody else can see it. For options, we have type radio which lets user pick only one circle from a group like Male or Female, and type checkbox which makes square boxes where you can tick multiple hobbies at once. Finally, type submit makes the main button which we click to send the data.

There are other elements also. If you want user to type a long message like an address, input box is too small, so we use textarea tag which makes a big box. If you want to show a dropdown list where users can choose their state, we use select tag along with option tags. We also use label tag to write text next to our input boxes so user knows what to type. To check if users have entered correct data, we use form validation. HTML5 gives some easy built-in options like required attribute. If you put this word inside any input tag, the user cannot leave that box empty. We also have min and max for number limits, and placeholder for hint text. We should connect our label tag to the input box using matching id attributes so screen readers can easily announce the box name for blind users.

Answer 5

Putting text content is the most basic work when making a website with HTML. To make the text look proper instead of a giant messy block, HTML gives us different tags. When we want to write a normal paragraph, we put it inside p tag. For titles and headers, HTML has six heading tags from h1 to h6. The h1 tag makes text biggest and boldest, and it is used for the main topic title, while h2 to h6 get smaller for sub-headings. Using these headings correctly helps the user to understand the page structure and read things smoothly.

In old HTML versions, if we wanted to change text alignment, we used an attribute called align inside the tags. For example, writing align center would move the text to the middle of screen, and we can also use left or right. Nowadays, doing alignment with HTML attributes is old style because everyone uses CSS for design, but for college students who are just starting to learn basic coding, using this align attribute is still very common. We also use tags like b for making text bold, i for italics, and br to break the line and go to next line manually.

To make web pages look colorful, we add images using img tag. This tag is an empty tag, it does not have any closing tag. It needs two main attributes to work. First is src attribute which means source; here we must type the exact name or folder path where our image file is stored on the computer. Second attribute is alt which means alternate text. Inside alt, we write a short text describing the photo. If user internet is slow and image does not load, the browser will show this alt text so user knows what was there. Also, reading software reads this text for blind people.

One big problem with images is file size. If you take a photo from mobile camera and upload it directly, the size will be very huge like 4MB. This will make the website open very slowly. So image compression is very important. This means we should change image dimensions to fit our page layout and make it small in kilobytes using some tool without making it look completely blurry. We should use old standard files like JPEG or PNG. Hyperlinks are what connect different pages together, so we can click and jump to another website address. We make links using anchor tag with href attribute to hold the website link. We should always write meaningful text inside the link like Download Syllabus instead of Click Here.

Answer 6

JavaScript is a coding language we use to make our web pages live and interactive. While HTML makes the page structure and CSS gives colors, JavaScript brings the actual working logic. It helps us to handle things like calculations, pop-up alerts, button animations, and changing text on the page without reloading the whole browser tab. Because of JavaScript, modern websites work like software apps instead of just flat reading pages.

When writing JavaScript code, we have to follow some basic syntax rules. First thing is JavaScript is case-sensitive. This means if you make a variable named myValue in small letters, and later you call it with capital letter, it will give an error saying variable not found. Both are totally different. Second rule is that every line usually ends with a semicolon. Even though modern browsers still run the code if you forget it, it is a good habit to put semicolons to avoid random bugs. Third, we use comments to write notes for ourselves. We use double slash for single-line notes, and star symbols for writing long paragraphs of notes that the browser should ignore.

JavaScript has different data types to store values inside variables. The most common is Number data type, which holds all mathematical figures like whole numbers like 40 or decimal values like 10.5. Next is String data type, which is used for writing plain text; we must always put string text inside quotes, like "BCA Code". Then we have Boolean data type, which can hold only two values: true or false. If we just create a variable but don't give any value to it, JavaScript sets it as Undefined. For big data, we use Arrays to make a list of items inside brackets, and Objects to store data in key and value format.

Operators are used for doing math and building logic. We have Arithmetic operators like plus, minus, multiply, divide for normal math calculations. We use single equal sign as Assignment operator to put a value into a variable. For comparing things, we use comparison symbols like double equal to check if values are equal, and triple equal which is strict because it checks both value and data type are same. We also use less than and greater than signs. If we want to check multiple conditions together, we combine them using logical items like AND and OR. JSON is basically just a short text format used for storing and sharing data values. The look of JSON is almost same to a normal JavaScript object. Inside our JavaScript code, we use parse method to convert text string into object, and stringify method to convert object into text string.